

Invariant measures for the nonlinear stochastic heat equation with no drift term

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Abstract

This paper deals with the long term behavior of the solution to the nonlinear stochastic heat equation $\frac{\partial u}{\partial t} - \frac{1}{2}\Delta u = b(u)\dot{W}$, where b is assumed to be a globally Lipschitz continuous function and the noise $\dot{W}$ is a centered and spatially homogeneous Gaussian noise that is white in time. We identify a set of nearly optimal conditions on the initial data, the correlation measure of the noise, and the weight function ρ , which together guarantee the existence of an invariant measure in the weighted space $L^2_\rho(\mathbb{R}^d)$. In particular, our result covers the *parabolic Anderson model* (i.e., the case when $b(u) = \lambda u$) starting from the Dirac delta measure.

Keywords: Stochastic heat equation, parabolic Anderson model, invariant measure, Dirac delta initial condition, weighted L^2 space, Matérn class of correlation functions, Bessel kernel.

MSC Classification: 60H15 , 35H60 , 37L40

1 Introduction

In this paper, we study the following *nonlinear stochastic heat equation* (SHE):

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) - \frac{1}{2} \Delta u(t, x) = b(x, u(t, x)) \dot{W}(t, x) & x \in \mathbb{R}^d, t > 0, \\ u(0, \cdot) = \mu(\cdot). \end{cases} \quad (1)$$

The noise, $\dot{W}(t, x)$, is a centered Gaussian noise that is white in time and homogeneously colored in space defined on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with the natural filtration $\{\mathcal{F}_t\}_{t \geq 0}$ generated by the noise. Its covariance structure is given by

$$J(\psi, \phi) := \mathbb{E}[W(\psi) W(\phi)] = \int_0^\infty ds \int_{\mathbb{R}^d} \Gamma(dx) (\psi(s, \cdot) * \tilde{\phi}(s, \cdot))(x), \quad (2)$$

where ψ and ϕ are continuous and rapidly decreasing functions, $\tilde{\phi}(x) := \phi(-x)$, “ $*$ ” refers to the convolution in the spatial variable, and Γ is a nonnegative and nonnegative definite tempered measure on $\mathbb{R}^d$ that is commonly referred to as the *correlation measure*. The Fourier transform of Γ (in the generalized sense) is also a nonnegative and nonnegative definite tempered measure, which is usually called the *spectral measure* and is denoted by $\hat{f}(d\xi)$ (see (21) for the convention of Fourier transform). Moreover, in the case where Γ has a density f , namely, $\Gamma(dx) = f(x)dx$, we write $\hat{f}(d\xi)$ as $\hat{f}(\xi)d\xi$.

The initial condition, μ , is a deterministic, locally finite, regular, signed Borel measure that satisfies the following integrability condition at infinity:

$$\int_{\mathbb{R}^d} \exp(-ax^2) \mu(dx) < \infty \quad \text{for all } a > 0, \quad (3)$$

where $\mu = \mu_+ + \mu_-$ and $\mu = \mu_+ - \mu_-$ refers to the *Hahn decomposition* of the measure μ . Initial conditions of this type, introduced in [1] and further explored in [2, 3], are called *rough initial conditions*.

The function $b(x, u)$ is uniformly bounded in the first variable and Lipschitz continuous in the second variable, i.e., for some constants $L_b > 0$ and $L_0 \geq 0$,

$$b(x, u) - b(x, v) \leq L_b u - v \quad \text{and} \quad b(x, 0) \leq L_0 \quad \text{for all } u, v \in \mathbb{R} \text{ and } x \in \mathbb{R}^d. \quad (4)$$

In particular, our assumption allows the linear case $b(x, u) = \lambda u$, which is usually referred to as the *parabolic Anderson model* (PAM) [4].

The SPDE (1) is understood in its *mild form*:

$$u(t, x) = J_0(t, x; \mu) + \int_0^t \int_{\mathbb{R}^d} b(y, u(s, y)) G(t - s, x - y) W(ds, dy), \quad (5)$$

where $G(t, x) = (2\pi t)^{-d/2} \exp\left(- (2t)^{-1} x^2\right)$ is the heat kernel,

$$J_0(t, x) = J_0(t, x; \mu) := (G(t, \cdot) * \mu)(x) = \int_{\mathbb{R}^d} G(t, x - y) \mu(dy) \quad (6)$$

is the solution to the homogeneous equation, and the stochastic integral is the *Walsh integral*. We refer the interested readers to [2, 5–7] for more details of this setup.

The aim of this paper is to investigate the conditions required to guarantee the existence of an invariant measure for the solution to (1), which is a crucial step for the study of the ergodicity of the system that requires the corresponding uniqueness. We direct the interested readers to [8–10] for more details about the invariant measure, its existence/uniqueness, and the ergodicity of the system. The general procedure for finding the invariant measure, especially in the setting of (1), has been laid out by Tessitore and Zabczyk [11], which involves two parts: first one needs to show that the laws of the solution to (1) form a family of *Markovian transition functions* on some Hilbert space, H , and the corresponding *Markovian semigroup* is Feller; and second one needs to establish that the moments of solution are bounded in time (see (8) below). For the second point, it requires some substantial work (see Theorems 1 and 3 below). On the other hand, the first point has been shown to be true for our case of interest (see, e.g., [10, Chapter 9]) with the following weighted $L^2(\mathbb{R}^d)$ space as our underlying Hilbert space as in [11]:

Definition 1 ([11]) A function $\rho : \mathbb{R}^d \mapsto \mathbb{R}$ is called an admissible weight function if it is a strictly positive, bounded, continuous, and $L^1(\mathbb{R}^d)$ -integrable function such that for all $T > 0$, there exists a constant $C_\rho(T)$ such that

$$(G(t, \cdot) * \rho(\cdot))(x) \leq C_\rho(T) \rho(x) \quad \text{for all } t \in [0, T] \text{ and } x \in \mathbb{R}^d. \quad (7)$$

Moreover, we denote by $L_\rho^2(\mathbb{R}^d)$ the corresponding Hilbert space of ρ -weighted square integrable functions, and we use $\langle \cdot, \cdot \rangle_\rho$ and $\|\cdot\|_\rho$ to denote the inner product and norm on $L_\rho^2(\mathbb{R}^d)$:

$$\langle f, g \rangle_\rho := \int_{\mathbb{R}^d} f(x)g(x)\rho(x)dx \quad \text{and} \quad \|f\|_\rho := \int_{\mathbb{R}^d} f(x)^2\rho(x)dx.$$

Accordingly, we will prove the existence of the invariant measure following the same strategy as in [11]. Let $\mathcal{L}(u(t, \cdot; \mu))$ denote the law of $u(t, \cdot)$ starting from μ at $t = 0$. We will first establish the tightness of $\{\mathcal{L}(u(t, \cdot; \mu))\}_{t \geq t_0}$ for some $t_0 \geq 0$. A critical step in obtaining this tightness result is to show that the following moment is uniformly bounded in time (see Theorem 3):

$$\sup_{t \geq 0} \mathbb{E} \left(\|u(t, \cdot)\|_\rho^2 \right) < \infty. \quad (8)$$

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Then we will apply the *Krylov-Bogoliubov theorem* (see, e.g., [10, Theorem 11.7]) to construct an invariant measure via

$$\eta(A) = \lim_{n \rightarrow \infty} \frac{1}{T_n} \int_{t_0}^{T_n+t_0} \mathcal{L}(u(t, \cdot; \mu))(A) dt, \quad (9)$$

for some sequence $\{T_n\}_{n \geq 1}$ with $T_n \uparrow \infty$.

In the literature, the existence of an invariant measure for the stochastic heat equation is more commonly studied with a drift term; we will postpone a brief review of this case to Section ???. In contrast, the existence of an invariant measure under the setting of equation (1) has been much less studied. Tessitore and Zabczyk [11] established the existence of an invariant measure for the case where the spatial domain is the whole space $\mathbb{R}^d$, the diffusion term, $b(x, u)$, is globally Lipschitz in the second variable, uniformly bounded in the first variable (see (4)) and there is no additional negative drift term to help. Recently, in the linear case, that is, the case when $b(u) = \lambda u$, Gu and Li [12] established the weak convergence of $u(t, \cdot)$ to $Z(\cdot)$ in $C(\mathbb{R}^d)$ with $Z(\cdot)$ being a stationary random field. The convergence in [12] relies on the fact that the initial condition is to be stationary, i.e., $\mu(dx) = dx$; see Remark 3.6 *ibid.* for some perturbations that can be made on the constant initial condition. Similar stationary limit has also been obtained in Dunlap *et al* [13]. The current paper follows largely the work by Tessitore and Zabczyk [11]. The major challenge is to identify the right conditions so that the probability moments of the solution are bounded in time (see (8)). The solution to (1) is usually *intermittent*, namely, its moments possess a certain exponential growth in t ; see, e.g., [4, 14]. For that reason, one has to impose some additional assumptions on the initial conditions, the noise, the coefficients of (1), and the weight function ρ in order to control the growth of the moments in $L^2((0, \infty); L^2_\rho(\mathbb{R}^d))$. The point-wise moment formulas obtained in [2, 3] play an important role in this context.

Here we emphasize that we study the invariant measure using the Walsh random field approach [7], whereas such studies are mostly carried out under the framework of the stochastic evolution in Hilbert spaces [10]. Even though both theories are equivalent (see [15]), the differences in many technical aspects are still substantial. As the random field approach often produces results that are more explicit, we try to use this approach to obtain more precise conditions for the existence of an invariant measure. *Firstly*, for the initial conditions, the results in [11] allow for bounded $L^2_\rho(\mathbb{R}^d)$ functions, although the authors proved their main result—Theorem 3.3 *ibid.*—only for the constant one initial condition. Here we give the precise conditions on the initial condition (see (17) below) which allows a wider class of data, including unbounded functions and measures such as the Dirac delta measure (see Examples ?? and ??). Note that the Dirac delta initial measure plays a prominent role in the study of the stochastic heat equation; see, e.g., [16]. *Secondly*, regarding the noise, we identify the right conditions, (10a) and (13), or equivalently (19), on the noise to ensure the existence of an invariant measure. Our conditions are explicit, easy

to verify, and in a compact form (see (19)). Although they are not necessary and sufficient conditions, we believe these conditions are towards to the optimal conditions; see Remark 2. Note that the correlation function in Gu and Li [12] is assumed to be a smooth function with compact support. The comparisons of our conditions with those obtained by Tessitore and Zabczyk [11] are given in Section ?? . *Thirdly*, our proof relies on a factorization representation for the solution $u(t, x)$ to (1) (see Lemma ??), which is obtained under the random field framework, whereas such factorization lemma has been widely used in the framework of the stochastic evolution equation in Hilbert spaces; see Section ?? for more details. *Finally*, we point out a miscellany of results in Section ??, which may have independent interest.

Now we are ready to motivate the conditions that we use and present the main results.

1.1 Main results

As mentioned earlier, in order to have moments bounded in time as in (8), one should better first identify the sharp conditions under which the second moments as a function of t , namely $t \mapsto \mathbb{E} (u(t, x)^2)$, with x fixed, are bounded. This question has been answered in [2, Theorem 1.3 and Lemma 2.5], where necessary and sufficient conditions are given. More precisely, to have the second moment bounded in time with x fixed, one needs to have the spatial dimension $d \geq 3$, and in addition, the spectral measure $\hat{f}$ and Lipschitz constant L_b of $b(\cdot)$ need to satisfy the following two conditions:

$$\Upsilon(0) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\xi^2} < \infty \quad (10a)$$

and

$$64L_b^2 < \frac{1}{2\Upsilon(0)} . \quad (10b)$$

These two conditions will guarantee the existence of the following non-empty open interval:

$$(2^7 L_b^2 \Upsilon(0), 1) \neq \emptyset. \quad (11)$$

Note that condition (10a) is a strengthened version of *Dalang's condition*:

$$\Upsilon(\beta) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\beta + \xi^2} < \infty, \quad \text{for some (and hence) all } \beta > 0. \quad (12)$$

It turns out that besides condition (10a) we need one more condition on the spectral measure $\hat{f}$, which strengthens Dalang's condition (12) in a different

way: that for some $\alpha \in (0, 1]$,

$$\Upsilon_\alpha(\beta) := (2\pi)^{-d} \int_{\mathbb{R}^d} \frac{\widehat{f}(\mathrm{d}\xi)}{(\beta + \xi^2)^{1-\alpha}} < \infty \quad \text{for some (hence all) } \beta > 0. \quad (13)$$

In some literature, condition (13) is called the *strengthened Dalang's condition*. These three conditions are illustrated in Figure 1.

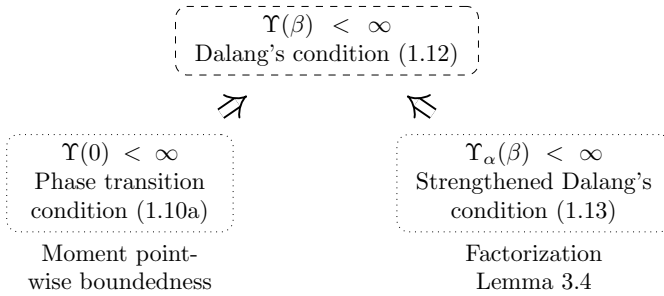


Fig. 1 Two conditions—(10a) and (13)—on $\widehat{f}$ needed for the main result — Theorem 3.

Remark 1 Dalang's condition (12) assures the existence and uniqueness of an $L^2(\Omega)$ -continuous solution; see [6]. But Dalang's condition (12) alone is sometimes too weak to derive some useful properties. Instead, the strengthened Dalang's condition (13) is widely assumed in the literature. For example, condition (13) is used to derive the Hölder regularity of the solution; see [3, Theorem 1.8] or [17] for the random field approach and Theorem 2.3 of [18] and Theorem 4.4 of [11] for the infinite-dimensional SDE approach. One may also check the densely blowup solution obtained in [19] where Dalang's condition (12) is satisfied but condition (13) fails. As another example, condition (13) is also required in order to derive the sample path comparison principle in [3]. As shown in Lemma ?? below, condition (13) guarantees a factorization of the solution, which plays a key role in the tightness argument in the proof of the main result – Theorem 3.

We are now ready to state our two main results of the paper.

Theorem 1 *Let $u(t, x; \mu)$ be the solution to (1) starting from μ which satisfies (3). Assume that*

- (i) $\rho : \mathbb{R}^d \rightarrow \mathbb{R}_+$ is a nonnegative $L^1(\mathbb{R}^d)$ function;
- (ii) for all $t > 0$, the initial condition μ satisfies $\mathcal{G}_\rho(t; \mu) < \infty$ where

$$\mathcal{G}_\rho(t; \mu) := \int_{\mathbb{R}^d} J_0^2(t, x; \mu) \rho(x) \, \mathrm{d}x; \quad (14)$$

(iii) the spectral measure $\hat{f}$ and the Lipschitz constant L_b satisfy the two conditions in (10).

Then there exists a unique $L^2(\Omega)$ -continuous solution $u(t, x)$ such that for some constant $C > 0$, which does not depend on t , the following holds:

$$\mathbb{E} \left(\|u(t, \cdot; \mu)\|_\rho^2 \right) \leq C \mathcal{G}_\rho(t; \mu^*) < \infty, \quad \text{for any } t > 0, \quad (15)$$

where $\mu^* := 1 + \mu$.

This theorem will be proved in Section ?? . We now state and prove a corollary which shows that the solution to (1) starting from an $L_\rho^2(\mathbb{R}^d)$ initial condition will almost surely be in $L_\rho^2(\mathbb{R}^d)$ for all $t > 0$.

Corollary 2 *Under the same assumptions of Theorem 1, if in addition ρ is admissible (see Definition 1), then the solution $u(t, \cdot; \zeta)$ to (1) almost surely exists in $L_\rho^2(\mathbb{R}^d)$ for all $t > 0$, whenever the initial condition is also in $L_\rho^2(\mathbb{R}^d)$, i.e., $\zeta \in L_\rho^2(\mathbb{R}^d)$.*

Proof Choose and fix an arbitrary $\zeta \in L_\rho^2(\mathbb{R}^d)$ and set $\zeta^* = 1 + \zeta$. It is clear that $\zeta^* \in L_\rho^2(\mathbb{R}^d)$. By (ii) of Theorem 1, it suffices to show the finiteness of $\mathcal{G}_\rho(t; \zeta^*)$ for all $t > 0$. Indeed, by Hölder's inequality, $\mathcal{G}_\rho(t; \zeta^*)$ is equal to

$$\int_{\mathbb{R}^d} \left(\int_{\mathbb{R}^d} G(t, x - y) \zeta^*(y) dy \right)^2 \rho(x) dx \leq \iint_{\mathbb{R}^{2d}} G(t, x - y) \zeta^*(y)^2 \rho(x) dy dx.$$

Now for any $t > 0$, choose $T > t$ and let $C_\rho(T)$ be as in (7). Then,

$$\iint_{\mathbb{R}^{2d}} G(t, x - y) \zeta^*(y)^2 \rho(x) dy dx = \int_{\mathbb{R}^d} (G(t, \cdot) * \rho)(y) \zeta(y)^2 dy \leq C_\rho(T) \|\zeta^*\|_\rho^2 < \infty.$$

Since t and T are arbitrary, this completes the proof of the corollary. $\square$

Theorem 3 *Let $u(t, x)$ be the solution to (1) starting from μ and let ρ be an admissible weight function. Assume that*

(i) *there exists another admissible weight $\tilde{\rho}$ such that*

$$\int_{\mathbb{R}^d} \frac{\rho(x)}{\tilde{\rho}(x)} dx < \infty; \quad (16)$$

(ii) *the weight function $\tilde{\rho}$ and the initial condition satisfy the following condition:*

$$\limsup_{t>0} \mathcal{G}_{\tilde{\rho}}(t; \mu) < \infty; \quad (17)$$

(iii) *the spectral measure $\hat{f}$ and the Lipschitz constant L_b satisfy the two conditions in (10);*

(iv) *for some $\alpha \in \left(2^7 \Upsilon(0) L_b^2, 1 \right)$ (see (11)), the spectral measure $\hat{f}$ satisfies the strengthened Dalang's condition (13).*

Then we have that

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(a) for any $\tau > 0$, the sequence of laws of $\{\mathcal{L}u(t, \cdot; \mu)\}_{t \geq \tau}$ is tight, i.e., for any $\epsilon \in (0, 1)$, there exists a compact set $\mathcal{K} \subset L^2_\rho(\mathbb{R}^d)$ such that

$$\mathcal{L}u(t, \cdot; \mu)(\mathcal{K}) \geq 1 - \epsilon, \quad \text{for all } t \geq \tau > 0; \quad (18)$$

(b) there exists an invariant measure for the laws $\{\mathcal{L}u(t, \cdot; \mu)\}_{t > 0}$ in $L^2_\rho(\mathbb{R}^d)$.

This theorem will be proved in Section ??.

Remark 2 The conditions on the spectral measure $\hat{f}$ for the existence of an invariant measure are given by conditions (iii) and (iv) of Theorem 3, which can be summarized in the following more compact form (thanks to Lemma ?? below):

$$\int_{\mathbb{R}^d} \frac{\hat{f}(d\xi)}{\xi^2 \wedge \xi^{2(1-\alpha)}} < \infty, \quad \text{for some } \alpha \in \left(2^7 \Upsilon(0) L_b^2, 1\right). \quad (19)$$

Even though condition (19) is not the necessary and sufficient condition for the existence of an invariant measure, it is almost the optimal condition. In particular, as proved in [2], if $\Upsilon(0) = \infty$, namely, condition (iii) of Theorem 3 fails, then no matter how small the level of the noise is, the solution is always intermittent, i.e., the second moment Lyapunov exponent of the solution is always strictly positive. Note that intermittency is another way to characterize the chaotic behavior of the underlying system that is far from equilibrium.

1.2 Outline and notation

The paper is organized as follows: we first prove Theorem 1 in Section ?. Then in Section ?, we study the factorization lemma and explore the relations of various conditions on the spectral measure $\hat{f}$. Then we proceed to prove Theorem 3 in Section ?. Finally, in Section ? we make some further discussion on the main results and present various examples. In particular, in Section ?, we give a brief review of the problem of finding invariant measures for the SHE with a drift term; in Section ?, we compare our conditions on the spectral density with those obtained by Tessitore and Zabczyk [11]; in Section ?, we show that our results could include a wider class of initial conditions; in Section ?, we carry out some explicit computations for the Bessel and related kernels as the correlation functions; finally, in Section ?, we give a few examples of the admissible weight functions.

We conclude this Introduction by introducing some notation and formulas that we use throughout the paper. We will use $\|X\|_p$ to denote the $L^p(\Omega)$ norm, namely, $\|X\|_p = (\mathbb{E}(X^p))^{1/p}$. We will also use the following factorization property of the heat kernel,

$$G(t, x)G(s, y) = G\left(\frac{ts}{t+s}, \frac{sx+ty}{t+s}\right) G(t+s, x-y), \quad (20)$$

which can be easily verified and has been used extensively and critically in [1–3]. Next, we will need the following spherical coordinate integration formula:

$$\int_{\mathbb{R}^d} f(x) dx = \sigma(\mathbb{S}^{d-1}) \int_0^\infty f(r) r^{d-1} dr,$$

where $\sigma(\mathbb{S}^{d-1}) = 2\pi^{d/2}/\Gamma(d/2)$ and $\Gamma(x)$ denotes the Gamma function. We use “ $\sim$ ” to denote the standard asymptotic equivalent relation. Lastly, the convention of Fourier transform is given by (see Remark ??)

$$\widehat{\phi}(\xi) = \mathcal{F}\phi(\xi) := \int_{\mathbb{R}^d} e^{-ix \cdot \xi} \phi(x) dx \quad \text{and} \quad \mathcal{F}^{-1}\psi(x) := (2\pi)^{-d} \int_{\mathbb{R}^d} e^{ix \cdot \xi} \psi(\xi) d\xi. \quad (21)$$

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Acknowledgments. L.C. thanks Sandra Cerrai for some helpful discussions and for pointing out the reference [20] during the conference – *Frontier Probability Days 2018*. L.C. also thanks Yu Gu for some helpful discussions regarding the stationary limit obtained in [12] during the *Workshop: Stochastic PDEs & Related Topics*, Nov. 14–16, 2022 at Brin Mathematics Research Center, University of Maryland, and for pointing out the reference [13].

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